

Create:

```
mysql> CREATE DATABASE HospitalDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE HospitalDB;
Database changed
mysql> CREATE TABLE Doctor (
->   DoctorID INT PRIMARY KEY,
->   DoctorName VARCHAR(50) NOT NULL,
->   Specialization VARCHAR(50) UNIQUE
-> );
Query OK, 0 rows affected (0.08 sec)

mysql> CREATE TABLE Patient (
->   PatientID INT PRIMARY KEY,
->   PatientName VARCHAR(50) NOT NULL,
->   Age INT CHECK (Age >= 0),
->   Gender CHAR(1) DEFAULT 'M',
->   Phone VARCHAR(10) UNIQUE,
->   DoctorID INT,
->   FOREIGN KEY (DoctorID) REFERENCES Doctor(DoctorID)
-> );
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> desc table Doctor;
```

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	Doctor	NULL	ALL	NULL	NULL	NULL	NULL	1	100.00	NULL

1 row in set, 1 warning (0.01 sec)

```
mysql> Desc Doctor;
```

Field	Type	Null	Key	Default	Extra
DoctorID	int	NO	PRI	NULL	
DoctorName	varchar(50)	NO		NULL	
Specialization	varchar(50)	YES	UNI	NULL	

```
mysql> Desc Patient;
```

Field	Type	Null	Key	Default	Extra
PatientID	int	NO	PRI	NULL	
PatientName	varchar(50)	NO		NULL	
Age	int	YES		NULL	
Gender	char(1)	YES		M	
Phone	varchar(10)	YES	UNI	NULL	
DoctorID	int	YES	MUL	NULL	

6 rows in set (0.00 sec)

Insert:

```
mysql> INSERT INTO Doctor
-> VALUES
-> (1, 'Dr. Kumar', 'Cardiology'),
-> (2, 'Dr. Priya', 'Neurology');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Patient
-> VALUES
-> (101, 'Arun', 25, 'M', '9876543210', 1),
-> (102, 'Divya', 30, 'F', '9876543211', 2);
Query OK, 2 rows affected (0.02 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> select * from Doctor;
+-----+-----+-----+
| DoctorID | DoctorName | Specialization |
+-----+-----+-----+
| 1 | Dr. Kumar | Cardiology |
| 2 | Dr. Priya | Neurology |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> select * from Patient;
+-----+-----+-----+-----+-----+-----+
| PatientID | PatientName | Age | Gender | Phone | DoctorID |
+-----+-----+-----+-----+-----+-----+
| 101 | Arun | 25 | M | 9876543210 | 1 |
| 102 | Divya | 30 | F | 9876543211 | 2 |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Constraint:

```

mysql> INSERT INTO Patient
-> VALUES
-> (101, 'Ravi', 28, 'M', '9876543212', 1);
ERROR 1062 (23000): Duplicate entry '101' for key 'patient.PRIMARY'
mysql> INSERT INTO Patient
-> VALUES
-> (103, 'Kiran', 22, 'M', '9876543210', 1);
ERROR 1062 (23000): Duplicate entry '9876543210' for key 'patient.Phone'
mysql> INSERT INTO Patient
-> (PatientID, Age, Gender, Phone, DoctorID)
-> VALUES
-> (104, 24, 'M', '9876543213', 1);
ERROR 1364 (HY000): Field 'PatientName' doesn't have a default value
mysql> INSERT INTO Patient
-> VALUES
-> (105, 'Meena', -5, 'F', '9876543214', 1);
ERROR 3819 (HY000): Check constraint 'patient_chk_1' is violated.
mysql> INSERT INTO Patient
-> (PatientID, PatientName, Age, Phone, DoctorID)
-> VALUES
-> (106, 'Suresh', 40, '9876543215', 2);
Query OK, 1 row affected (0.02 sec)

```

```

mysql> SELECT * FROM Patient;
+-----+-----+-----+-----+-----+-----+
| PatientID | PatientName | Age | Gender | Phone | DoctorID |
+-----+-----+-----+-----+-----+-----+
| 101 | Arun | 25 | M | 9876543210 | 1 |
| 102 | Divya | 30 | F | 9876543211 | 2 |
| 106 | Suresh | 40 | M | 9876543215 | 2 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

```

```

mysql> INSERT INTO Patient
-> VALUES
-> (107, 'Ajay', 29, 'M', '9876543216', 5);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('hospitaldb'.patient, CONSTRAINT 'patient_ibfk_1' FOREIGN KEY ('DoctorID') REFERENCES 'doctor' ('DoctorID'))

```